2.1

1. The elements C and Se have the same electronegativity value, 2.55. Which of the following claims about the compound that forms from C and Se is most likely to be true?

(A) The carbon-to-selenium bond is unstable.

(B) The carbon-to-selenium bond is nonpolar covalent. ✓

(C) The compound has the empirical formula CSe.

(D) A molecule of the compound will have a partial negative charge on the carbon atom.

Answer B

Correct. The bonding in compounds between elements with the same (or only slightly different) electronegativity values will exhibit covalent bonding (that is, more or less equal sharing of electrons).

2. Two pure elements react to form a compound. One element is an alkali metal, X, and the other element is a halogen, Z. Which of the following is the most valid scientific claim that can be made about the compound?

(A) It has the formula XZ_2.

(B) It does not dissolve in water.

(C) It contains ionic bonds. ✓

(D) It contains covalent bonds.

Answer C

Correct. An alkali metal will readily lose an electron and a halogen will readily gain an electron to form an ionic compound with a $1{:}1$ ratio of X^+: Z^- ions.

3. Which of the following scientific claims about the bond in the molecular compound HF is most likely to be true?

(A) There is a partial negative charge on the H atom.

(B) Electrons are shared equally between the H and F atoms.

(C) The bond is extremely weak.

(D) The bond is highly polar. ✓

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Answer D

Correct. Fluorine is the most electronegative element, and the difference between the electronegativities of H and F is relatively large, resulting in a bond that is highly polar.

4.

Element	Electronegativity
H	2.1
C	2.5
S	2.5
F	4.0
Cl	3.0
Si	1.8

On the basis of the information above, which of the following arranges the binary compounds in order of increasing bond polarity?

(A) $CH_4 < SiCl_4 < SF_4$ ✓

(B) $CH_4 < SF_4 < SiCl_4$

(C) $SF_4 < CH_4 < SiCl_4$

(D) $SiCl_4 < SF_4 < CH_4$

5. Which of the following has the bonds arranged in order of decreasing polarity?

(A) H−F > N−F > F−F ✓

(B) H−I > H−Br > H−F

(C) O−N > O−S > O−Te

(D) Sb−I > Sb−Te > Sb−Cl

6. On the basis of electronegativity differences between atoms, which of the following scientific claims is the most accurate regarding the bonding in $\mathrm{Mg(NO_3)_2}(s)$?

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(A) There is polar covalent bonding between Mg atoms and N atoms.

(B) There is polar covalent bonding between Mg atoms and O atoms.

(C) There is ionic bonding between N^{5+} ions and O^{2-} ions.

(D) There is ionic bonding between Mg^{2+} ions and NO_3^- ions. ✓

Answer D

Correct. There is ionic bonding in solid $Mg(NO_3)_2$ due to coulombic attractions between Mg^{2+} ions and NO_3^- ions. (There is some polar covalent bonding between N and O within the NO_3^- ion.)

7. Which of the following bonds is most polar?

(A) F-F

(B) O-F

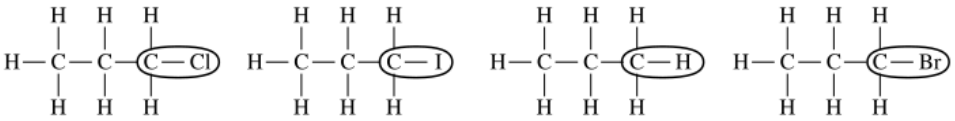
(C) O-I ✓

(D) O-Cl

Answer C

Correct. The polarity of bonds depends on the difference in the electronegativities of the atoms in the molecule; the greater the difference in electronegativity, the more polar the bond. Electronegativity generally decreases going down a column and from right to left across a row. Of the pairs given in the options, oxygen and iodine are farthest apart, have the greatest difference in electronegativity, and therefore form bonds that are the most polar.

8.



Which of the following bonds represented in the diagram above is the most polar?

(A) $C-Cl$ ✓

(B) $C-I$

(C) $C-H$

(D) $C-Br$

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Answer A

Correct. The $C-Cl$ bond is the most polar of the bonds in the responses because Cl is the most electronegative of Cl, I, H and Br. This can be inferred given that C and H have similar electronegativities and the periodic trend is for electronegativity to decrease from the top to the bottom of a column.

9. Which of the following compounds contains both ionic and covalent bonds?

(A) C_2H_5OH

(B) MgF_2

(C) H_2S

(D) NH_4Cl ✓

10. Which of the following claims about a binary compound composed of elements with the same electronegativity is most likely to be true?

(A) The compound has properties similar to those of both elements.

(B) The bonding in the compound is nonpolar covalent. ✓

(C) The boiling point of the compound is above $1000^\circ C$.

(D) The compound contains strong ionic bonds.

Answer B

Correct. Elements with the same electronegativity are more likely to share electrons and form a nonpolar covalent bond.

11. Which of the following substances has bonds with the greatest ionic character?

(A) CCl_4

(B) $CuCl_2$

(C) $CaCl_2$ ✓

(D) NCl_3

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Answer C

Correct. The degree of ionic character in a chemical bond is a function of the difference in electronegativities of the two atoms that form the bond. Among C, Cu, Ca, and N, the element with the lowest electronegativity is Ca as, it is to the left of the other elements on the periodic table. Thus, the difference between the electronegativity of Ca and the electronegativity of Cl is the largest difference, and the Ca – Cl bond has the greatest ionic character.

12.

$$PCl_3, Cl_2, CaCl_2$$

Which of the following correctly ranks the substances listed above in order of increasing ionic character in their bonding?

(A) Cl_2, PCl_3, $CaCl_2$ ✓

(B) Cl_2, $CaCl_2$, PCl_3

(C) $CaCl_2$, PCl_3, Cl_2

(D) $CaCl_2$, Cl_2, PCl_3

Answer A

Correct. The ionic character of a bond increases with increasing differences between the electronegativities of the bonded elements. The bond in Cl_2 is the least ionic because the two bonded atoms are identical. The bonds in $CaCl_2$ are the most ionic because the bonded atoms (Ca and Cl) have the greatest difference in their electronegativities, which can be inferred from their relative positions in the periodic table. Ca and Cl are on opposite sides of the periodic table, so the difference in their electronegativities is relatively large. The bonds in $CaCl_2$ are more ionic than those in PCl_3 because the difference in electronegativities between Ca and Cl is greater than that between P and Cl.

13. An atom of chlorine and an atom of which of the following elements will form a bond with the most ionic character?

(A) Germanium

(B) Carbon

(C) Copper

(D) Potassium ✓

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Answer D

Correct. The percent ionic character in a bond between two atoms increases with the difference in electronegativity between the bonded atoms. The electronegativity of elements varies with their position in the periodic table, increasing from left to right across a period and from bottom to top in a group. The greatest difference in electronegativity will be between chlorine and potassium because potassium is farther from chlorine on the periodic table than germanium, carbon, or copper are. Thus, the potassium-chlorine bond (K-Cl) will have the most ionic character.

14. Which of the following claims about a binary compound in which the bonding is ionic is most likely to be scientifically valid?

(A) Both elements in the compound are metals.

(B) The atomic masses of the elements in the compound are relatively small.

(C) There is equal sharing of electrons between the atoms of the elements in the compound.

(D) The electronegativity difference between the elements in the compound is relatively large. ✓

Answer D

Correct. A relatively large difference between the electronegativities of the two elements indicates that one element has a tendency to form cations and the other has a tendency to form anions.

15. Of the following compounds, which is the most ionic?

(A) $SiCl_4$

(B) BrCl

(C) PCl_3

(D) $CaCl_2$ ✓

16. Of the following single bonds, which is the LEAST polar?

(A) N—H

(B) H—F

(C) O—F ✓

(D) I—F